57. 12-Bit D/A Converter (R12DAa)

57.1 Overview

This MCU includes two channels of 12-bit D/A converter with an output buffer amplifier.

Table 57.1 lists the specifications of the 12-bit D/A converter and Figure 57.1 shows a block diagram of the 12-bit D/A converter.

Table 57.1 Specifications of 12-Bit D/A Converter

Item	Specifications
Resolution	12 bits
Output channels	Two channels
Measure against mutual interference between analog modules	Measure against interference between D/A and A/D conversion D/A converted data update timing is controlled by the 12-bit A/D converter synchronous D/A conversion enable signal from the 12-bit A/D converter (unit 1). Therefore, the degradation of A/D conversion accuracy due to interference is reduced by controlling the timing in which the 12-bit D/A converter inrush current occurs, with the enable signal.
Low power consumption function	Module stop state can be set.
Event link function (input)	DA0 conversion can be started when an event signal is input.
Output buffer amplifier control function	Buffered output (gain = 1) or unbuffered output can be selected.

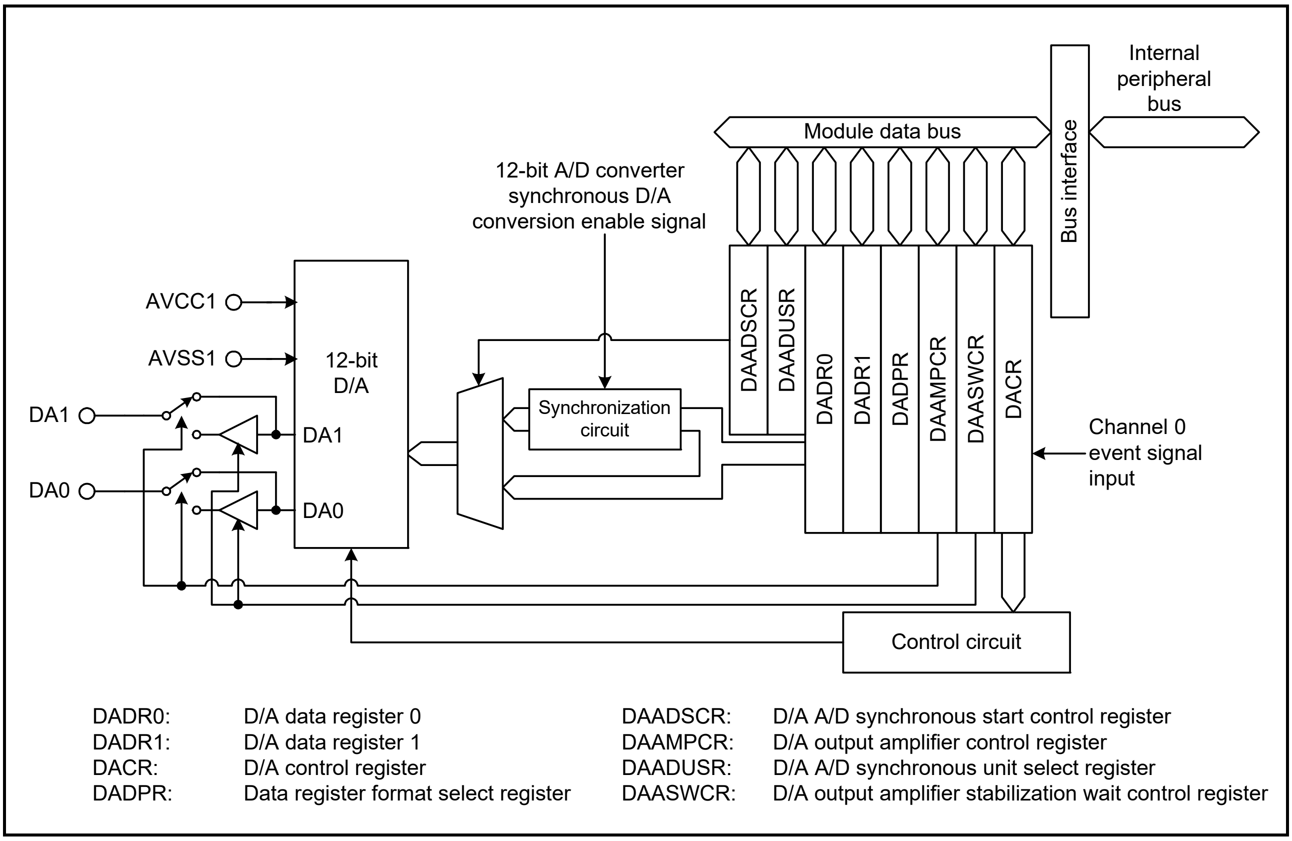

Figure 57.1 Block Diagram of 12-Bit D/A Converter

Table 57.2 lists the pin configuration of the 12-bit D/A converter.

Table 57.2 Pin Configuration of 12-Bit D/A Converter

Pin Name	I/O	Function
AVCC1	Input	Analog power supply pin and reference voltage supply pin
AVSS1	Input	Analog ground pin and reference voltage supply pin
DA0	Output	Channel 0 analog output pin
DA1	Output	Channel 1 analog output pin

57.2 Register Descriptions

57.2.1 D/A Data Register m (DADRm) (m = 0, 1)

Address(es): DA.DADR0 0008 8040h, DA.DADR1 0008 8042h

• DADPR.DPSEL bit = 0 (data is right-justified)

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

• DADPR.DPSEL bit = 1 (data is left-justified)

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
													—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The DADRm register is a 16-bit readable/writable register, which stores data to which D/A conversion is to be performed. Whenever an analog output is enabled, the values in DADRm are converted and output from the D/A converter.

12-bit data can be relocated by setting the DADPR.DPSEL bit.

Bits "—" are read as 0. The write value should be 0.

When the output buffer amplifier is used, see section 57.6.5, Initial Setting Procedure when the Output Buffer Amplifier is Used.

57.2.2 D/A Control Register (DACR)

Address(es): DA.DACR 0008 8044h

	b7	b6	b5	b4	b3	b2	b1	b0
	DAOE1	DAOE0	DAE	—	—	—	—	—
Value after reset:	0	0	0	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b5	DAE	D/A Enable*1	0: D/A conversion of channels 0 and 1 is controlled individually. 1: D/A conversion of channels 0 and 1 is enabled collectively.	R/W
b6	DAOE0	D/A Output Enable 0	0: Analog output of channel 0 (DA0) is disabled. 1: D/A conversion of channel 0 is enabled. Analog output of channel 0 (DA0) is enabled.	R/W
b7	DAOE1	D/A Output Enable 1	0: Analog output of channel 1 (DA1) is disabled. 1: D/A conversion of channel 1 is enabled. Analog output of channel 1 (DA1) is enabled.	R/W

Note 1. This bit controls D/A conversion in combination with the DAOEm bit (m = 0, 1). The DAOEm bit controls output of the results of conversion. For details, see Table 57.3.

Table 57.3 Controls of D/A Conversion

b5	b7	b6	
DAE	DAOE1	DAOE0	Description
0	0	0	D/A conversion and analog output pins (DA0, DA1) are disabled.*1
		1	D/A conversion of channel 0 is enabled. D/A conversion of channel 1 is disabled. Analog output of channel 0 (DA0) is enabled. Analog output of channel 1 (DA1) is disabled.*1
	1	0	D/A conversion of channel 0 is disabled. D/A conversion of channel 1 is enabled. Analog output of channel 0 (DA0) is disabled.*1 Analog output of channel 1 (DA1) is enabled.
		1	D/A conversion of channels 0 and 1 is enabled. Analog output of channels 0 and 1 (DA0, DA1) is enabled.
1	x	x	D/A conversion of channels 0 and 1 is enabled. Analog output of channels 0 and 1 (DA0, DA1) is enabled collectively.

x: Don't care

Note 1. When analog output is disabled, the analog output signal is placed in the Hi-Z state.

This register should be set when the DAADSCR.DAADST bit is 1 (measure against interference between D/A and A/D conversion is enabled) while the 12-bit A/D converter (unit 1) is halted (the ADCSR.ADST bit is 0). At that time, the software trigger should be selected for the 12-bit A/D converter (unit 1) trigger to securely stop the 12-bit A/D converter (unit 1).

DAE Bit (D/A Enable)

The DAE bit controls D/A conversion, amplifier operation, and analog output in combination with the DAOEm bit (m = 0, 1), the DAAMPCR.DAAMPm bit, and the DAASWCR.DAASWm bit. See Table 57.4 for details.
When the measure against an interference between D/A and A/D conversions is enabled (DAADSCR.DAADST = 1), set the DAE bit while the ADCSR.ADST bit in the 12-bit A/D converter (unit 1) is 0. At that time, the software trigger should be selected for the 12-bit A/D converter (unit 1) trigger to securely stop the 12-bit A/D converter (unit 1).

DAOEm Bit (D/A Output Enable m) (m = 0, 1)

The DAOEm bit controls D/A conversion, amplifier operation, and analog output in combination with the DAE bit, DAAMPCR.DAAMPm bit, and the DAASWCR.DAASWm bit. See Table 57.4 for details.
When both the DAOEm bit and DAE bit are 0, D/A conversion of channel m is not done and no conversion result is output.
When the measure against an interference between D/A and A/D conversions is enabled (DAADSCR.DAADST = 1), set the DAOEi bit while the ADCSR.ADST bit in the 12-bit A/D converter (unit 1) is 0. At that time, the software trigger should be selected for the 12-bit A/D converter (unit 1) trigger to securely stop the 12-bit A/D converter (unit 1).
The event link function can be used to set the DAOE0 bit to 1. The DAOE0 bit becomes 1 when the event specified by setting the ELSR16 register of the ELC occurs, and output of the D/A conversion results starts.

Table 57.4 D/A Conversion and Analog Output Control (m = 0, 1)

DACR		DAAMPCR	DAASWCR	D/A Conversion Operation of Channel m	Amplifier Operation of Channel m	Analog Output of Channel m
DAE	DAOEm	DAAMPm	DAASWm			
0	0	0	0	Disabled	Disabled	Hi-Z
			1	Disabled	Disabled	Hi-Z
		1	0	Disabled	Disabled	Hi-Z
			1	Disabled	Disabled	Hi-Z
	1	0	0	Enabled	Disabled	Unbuffered output
			1	Enabled	Disabled	Hi-Z
		1	0	Enabled	Enabled	Buffered output
			1	Enabled	Enabled	Hi-Z
1	0	0	0	Enabled	Disabled	Unbuffered output
			1	Enabled	Disabled	Hi-Z
		1	0	Enabled	Enabled	Buffered output
			1	Enabled	Enabled	Hi-Z
	1	0	0	Enabled	Disabled	Unbuffered output
			1	Enabled	Disabled	Hi-Z
		1	0	Enabled	Enabled	Buffered output
			1	Enabled	Enabled	Hi-Z

57.2.3 Data Register Format Select Register (DADPR)

Address(es): DA.DADPR 0008 8045h

	b7	b6	b5	b4	b3	b2	b1	b0
	DPSEL	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	DPSEL	Format Select	0: Data is right-justified. 1: Data is left-justified.	R/W

57.2.4 D/A A/D Synchronous Start Control Register (DAADSCR)

Address(es): DA.DAADSCR 0008 8046h

	b7	b6	b5	b4	b3	b2	b1	b0
	DAADST	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	DAADST	D/A A/D Synchronous Conversion	0: 12-bit D/A converter operation does not synchronize with 12-bit A/D converter (unit 1) operation. (measure against interference between D/A and A/D conversion is disabled) 1: 12-bit D/A converter operation synchronizes with 12-bit A/D converter (unit 1) operation. (measure against interference between D/A and A/D conversion is enabled)	R/W

As a measure against interference between D/A and A/D conversion, the DAADSCR register selects whether or not the timing for starting 12-bit D/A conversion is synchronized with the 12-bit A/D converter synchronous D/A conversion enable signal from the 12-bit A/D converter (unit 1).
This register should be set while the 12-bit A/D converter (unit 1) is halted (while the ADCSR.ADST bit is 0 after selecting software trigger as the 12-bit A/D converter (unit 1) trigger).
Unit 1 should be selected as the target unit of the 12-bit A/D converter before setting the DAADST bit to 1.
Set the DAADUSR.AMADSEL1 bit to 1 to select unit 1.

DAADST Bit (D/A A/D Synchronous Conversion)

Setting the DAADST bit to 0 allows the DADRm register value (m = 0, 1) to be converted into analog data at any time.
Setting the DAADST bit to 1 allows synchronous D/A conversion with the 12-bit A/D converter synchronous D/A conversion enable signal from the 12-bit A/D converter (unit 1). Therefore, even if the DADRm register value is modified, D/A conversion does not start until the 12-bit A/D converter (unit 1) completes A/D conversion.
Set this bit while the ADCSR.ADST bit is set to 0. At this time, the software trigger should be selected for the 12-bit A/D converter trigger to securely stop the 12-bit A/D converter (unit 1). Set the DAADUSR.AMADSEL1 bit to 1 before setting the DAADST bit to 1.
The event link function cannot be used when the DAADST bit is set to 1. Stop the event link function by setting the ELSR16 register of the ELC. The setting of the DAADST bit is common to channels 0 and 1 of the 12-bit D/A converter.

57.2.5 D/A Output Amplifier Control Register (DAAMPCR)

Address(es): DA.DAAMPCR 0008 8048h

	b7	b6	b5	b4	b3	b2	b1	b0
	DAAMP1	DAAMP0	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	DAAMP0	Amplifier Control 0	0: Output buffer amplifier of channel 0 is not used. 1: Output buffer amplifier of channel 0 is used.	R/W
b7	DAAMP1	Amplifier Control 1	0: Output buffer amplifier of channel 1 is not used. 1: Output buffer amplifier of channel 1 is used.	R/W

The DAAMPCR register selects D/A converter output with or without using the output buffer amplifier.

DAAMP0 Bit (Amplifier Control 0)

When the DAAMP0 bit is 0, the conversion result for the D/A converter channel 0 is output without using the output buffer amplifier. When the DAAMP0 bit is 1, the conversion result for the D/A converter channel 0 is output via the output buffer amplifier.

When both the DAE and DAOE0 bits are 0, the output buffer amplifier is disabled regardless of the setting of the DAAMP0 bit. See Table 57.4 for details.

DAAMP1 Bit (Amplifier Control 1)

When the DAAMP1 bit is 0, the conversion result for the D/A converter channel 1 is output without using the output buffer amplifier. When the DAAMP1 bit is 1, the conversion result for the D/A converter channel 1 is output via the output buffer amplifier.

When both the DAE and DAOE1 bits are 0, the output buffer amplifier is disabled regardless of the setting of the DAAMP1 bit. See Table 57.4 for details.

57.2.6 D/A Output Amplifier Stabilization Wait Control Register (DAASWCR)

Address(es) DA.DAASWCR 0008 805Ch

	b7	b6	b5	b4	b3	b2	b1	b0
	DAAS W1	DAAS W0	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5-b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	DAASW0	Channel 0 Output Amplifier Stabilization Waiting	0: A wait for stabilization of the channel 0 output buffer amplifier is released (output is enabled). 1: Wait for the channel 0 output buffer amplifier to become stable (the pin is Hi-Z).	R/W
b7	DAASW1	Channel 1 Output Amplifier Stabilization Waiting	0: A wait for stabilization of the channel 1 output buffer amplifier is released (output is enabled). 1: Wait for the channel 1 output buffer amplifier to become stable (the pin is Hi-Z).	R/W

The DAASWCR register is used to control the output buffer amplifier of the D/A converter during the initial setting when the output buffer amplifier is used.

DAASW0 Bit (Channel 0 Output Amplifier Stabilization Waiting)

When setting the DAASW0 bit to 1, the output buffer amplifier of D/A converter channel 0 enters into the stabilization waiting state and the DA0 pin becomes high impedance. When setting the DAASW0 bit to 0, the conversion result of the D/A converter channel 0 is output from the DA0 pin via the output buffer amplifier.
Set the DAASW0 bit to 1 when the DACR.DAE and DACR.DAOE0 bits are 0. After setting the DACR.DAE or DACR.DAOE0 bit to 1, wait for the required stabilization time and set the DAASW0 bit to 0. Refer to section 57.6.5, Initial Setting Procedure when the Output Buffer Amplifier is Used for details.

DAASW1 Bit (Channel 1 Output Amplifier Stabilization Waiting)

When setting the DAASW1 bit to 1, the output buffer amplifier of D/A converter channel 1 enters into the stabilization waiting state and the DA1 pin becomes high impedance. When setting the DAASW1 bit to 0, the conversion result of the D/A converter channel 1 is output from the DA1 pin via the output buffer amplifier.
Set the DAASW1 bit to 1 when the DACR.DAE and DACR.DAOE1 bits are 0. After setting the DACR.DAE or DACR.DAOE1 bit to 1, wait for the required stabilization time and set the DAASW1 bit to 0. Refer to section 57.6.5, Initial Setting Procedure when the Output Buffer Amplifier is Used for details.

57.2.7 D/A A/D Synchronous Unit Select Register (DAADUSR)

Address(es): DA.DAADUSR 0008 C5C0h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	AMADSEL1	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b1	AMADSEL1	A/D Unit 1 Select	0: Unit 1 is not selected. 1: Unit 1 is selected.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The DAADUSR register is used to select the target unit of the 12-bit A/D converter for D/A and A/D synchronous conversions. When setting the DAADSCR.DAADST bit to 1 for synchronous conversions, the target unit should be selected by this register in advance. Set the AMADSEL1 bit to 1 and select unit 1 as the target synchronous unit.

The DAADUSR register should be set when the ADCSR.ADST bit of the 12-bit A/D converter is set to 0 and the DAADSCR.DAADST bit is set to 0.

57.3 Operation

The 12-bit D/A converter includes D/A conversion circuits for two channels, each of which can operate independently. When the DACR.DAOEm bit (m = 0, 1) is set to 1, D/A converter is enabled and the conversion result is output.
An operation example of D/A conversion on channel 0 is shown below. Figure 57.2 shows the timing of this operation.

(1) Set the data for D/A conversion in the DADPR.DPSEL bit and the DADR0 register.

(2) Set the DACR.DAOE0 bit to 1 to start D/A conversion. The DA0 output settles to the voltage corresponding to the setting value after the conversion time tDCONV has elapsed. The DA0 output voltage is held at this level until the DADR0 register is updated or the DAOE0 bit is set to 0. The output voltage (reference) is expressed by the following formula:

$$\frac{\text{Value of DADRm register}}{4096} \times \text{AVCC1}$$

When the output buffer amplifier is used, the output voltage does not reach AVSS1 or AVCC1. Refer to section 63, Electrical Characteristics for the output voltage range.

(3) When the DADR0 register is updated, the conversion starts. The DA0 output settles at the new output voltage after the conversion time tDCONV has elapsed.
When the DAADSCR.DAADST bit is 1 (measure against interference between D/A and A/D conversion is enabled), it takes a maximum of one A/D conversion time for D/A conversion to start.

(4) When the DAOE0 bit is set to 0, analog output is disabled.

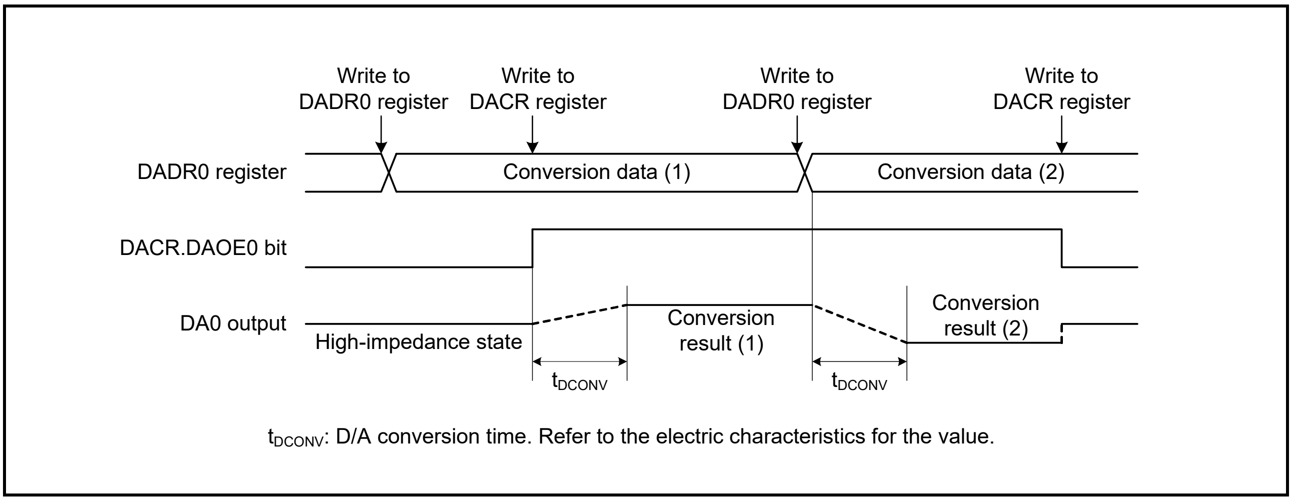

Figure 57.2 Example of 12-Bit D/A Converter Operation

57.3.1 Measure against Interference between D/A and A/D Conversion

When D/A conversion starts, an inrush current occurs to the 12-bit D/A converter. Since the same analog power supply is shared by the 12-bit D/A converter and 12-bit A/D converter (unit 1), the inrush current may interfere with the proper operation of the 12-bit A/D converter (unit 1).
With the DAADSCR.DAADST bit being 1, even if the DADRm register data (m = 0, 1) is modified during 12-bit A/D converter (unit 1) operation, D/A conversion does not start immediately but starts synchronously with A/D conversion completion. It takes a maximum of one A/D conversion time for the DADRm register data update to be reflected as the D/A conversion circuit input. Before reflection, the DADRm register value does not correspond to the analog output value.
When this function is enabled, it is impossible to check by any software means whether the DADRm register value has been D/A converted or not.
Even with the DAADSCR.DAADST bit being 1, when the DADRm register data is modified while the 12-bit A/D converter (unit 1) is halted, D/A conversion starts in one PCLKB cycle.
Figure 57.3 shows an example of channel 0 D/A conversion, in which the 12-bit D/A converter operates synchronously with the 12-bit A/D converter (unit 1).

(1) Confirm that the 12-bit A/D converter (unit 1) is halted. Set the DAADUSR.AMADSEL1 bit to 1.
(2) Confirm that the 12-bit A/D converter (unit 1) is halted. Set the DAADSCR.DAADST bit to 1.
(3) Confirm that the 12-bit A/D converter (unit 1) is halted. Set the DACR.DAOE0 bit to 1.
(4) Set the DADR0 register.

- If the 12-bit A/D conversion is halted (ADCSR.ADST bit = 0) when the DADR0 register is modified, D/A conversion starts in one PCLKB cycle.
- If the 12-bit A/D conversion is in progress (ADCSR.ADST bit = 1) when the DADR0 register is modified, D/A conversion starts upon A/D conversion completion. If the DADR0 register is modified twice during A/D conversion, the first update may not be converted.

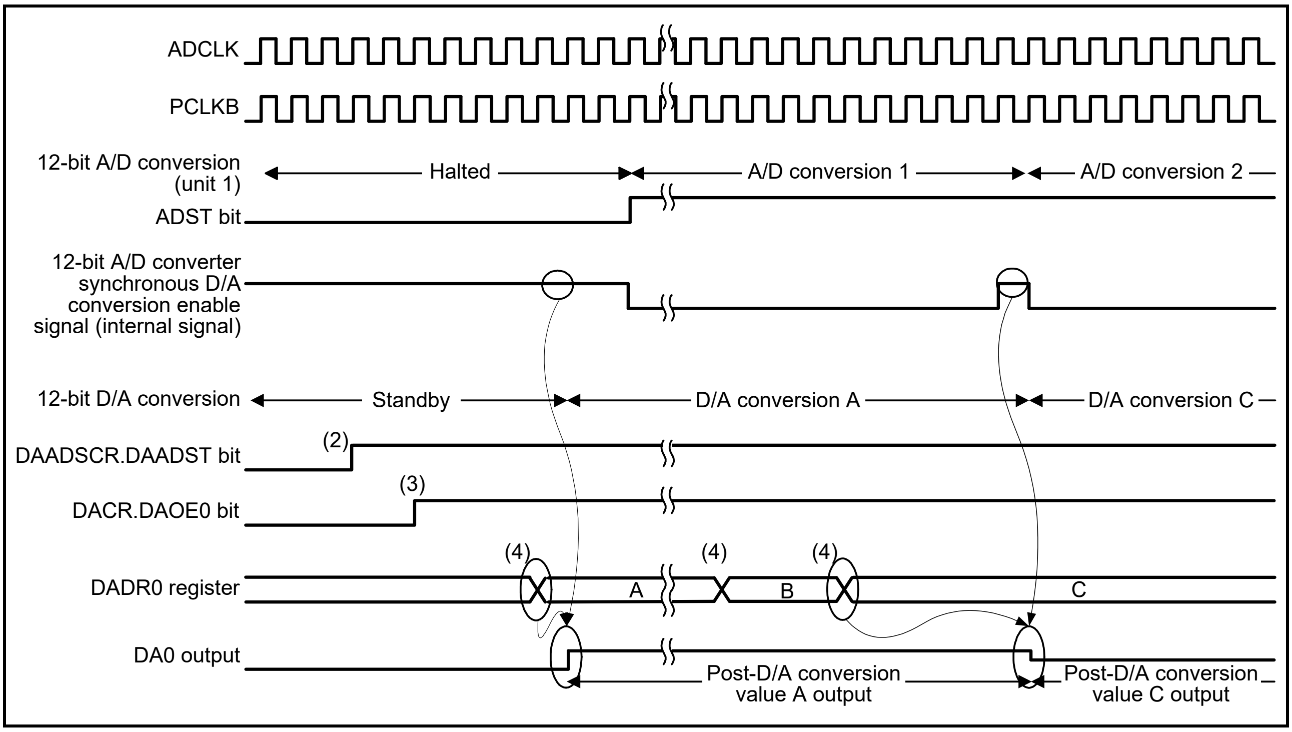

Figure 57.3 Example of Conversion When the 12-Bit D/A Converter is Synchronized with the 12-Bit A/D Converter (Unit 1)

57.4 Event Link Operation Setting Procedure

The event link operation procedure is described below.

(1) Set the DADPR.DPSEL bit and set the data for D/A conversion in the DADR0 register.
(2) Set the bit value of the ELSR16 setting event signal to link the ELSR16 register of the ELC.
(3) Set the ELCR.ELCON bit to 1. This procedure enables event link operation for all modules with the event link function selected.
(4) Set the event output source module to activate the event link. After the event is output from the module, the DACR.DAOE0 bit becomes 1, and D/A conversion on channel 0 starts.
(5) Set the ELSR16.ELS[7:0] bits to 0000 0000b to stop event link operation of 12-bit D/A converter channel 0. All event link operation is stopped when the ELCR.ELCON bit is set to 0.

57.5 Usage Notes on Event Link Operation

(1) When the event link function is used, do not use the output buffer amplifier.
(2) When the event link function is used, set the DACR.DAE bit to 0.
(3) When the event specified by the ELSR16 register is generated while the write cycle is performed to the DACR.DAOE0 bit, the write cycle is stopped, and the setting to 1 by the generated event takes precedence.
(4) Use of the event link function is prohibited when the DAADSCR.DAADST bit is set to 1 as the measure against an interfere between D/A and A/D conversions.

57.6 Usage Notes

57.6.1 Module Stop Function Setting

Operation of the 12-bit D/A converter can be disabled or enabled using the module stop control register. The initial setting is for operation of the 12-bit D/A converter to be stopped. Register access is enabled by releasing the module stop state. For details, refer to section 11, Low Power Consumption.

57.6.2 Operation of the D/A Converter in Module Stop State

When the MCU enters the module stop state with D/A conversion enabled, the D/A converter outputs are retained, and the analog power supply current is the same as during D/A conversion. If the analog power supply current has to be reduced in the module stop state, disable D/A conversion by setting the DACR.DAOE1, DAOE0, and DAE bits to 0.

57.6.3 Operation of the D/A Converter in Software Standby Mode

When the MCU enters software standby mode with D/A conversion enabled, the D/A converter outputs are retained, and the analog power supply current is the same as during D/A conversion. If the analog power supply current has to be reduced in software standby mode, disable D/A conversion by setting the DACR.DAOE1, DAOE0, and DAE bits to 0.

57.6.4 Note on Entering Deep Software Standby Mode

When the MCU enters deep software standby mode with D/A conversion enabled, the outputs of the D/A converter are placed in a high impedance state.

57.6.5 Initial Setting Procedure when the Output Buffer Amplifier is Used

When using the output buffer amplifier, enable the amplifier output in the following procedure. An example for channel 0 is described below.

(1) Confirm that the DACR.DAE and DACR.DAOE0 bits are 0.
(2) Write 0000h to the DADR0 register.
(3) Set the DAASWCR.DAASW0 bit to 1.
(4) Set the DAAMPCR.DAAMP0 bit to 1.
(5) Set the DACR.DAE or DACR.DAOE0 bit to 1. The output buffer amplifier starts the operation.
(6) Wait for at least 3 µs and then set the DAASWCR.DAASW0 bit to 0.
(7) Write a value to be converted in the DADR0 register.

While the output buffer amplifier is operating, setting the DACR.DAE and DACR.DAOE0 bits to 0 disables the output buffer amplifier. Repeat the procedure from (1) to (7) to use the output buffer amplifier again.

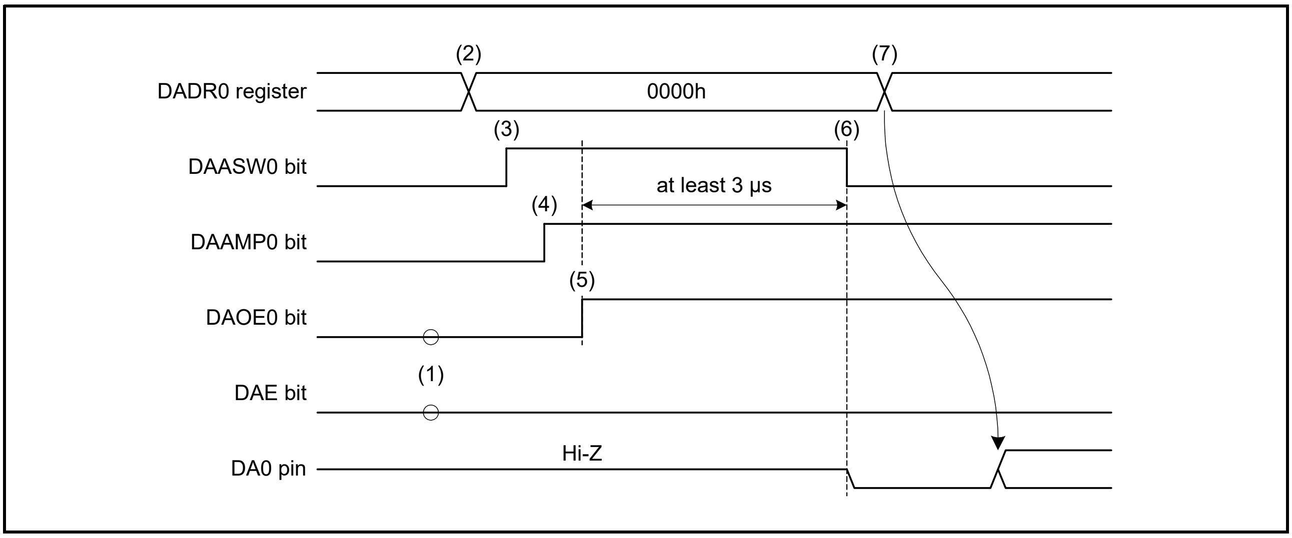

Figure 57.4 Initial Setting Procedure of the Output Buffer Amplifier

57.6.6 Note on Usage When Measure against Interference between D/A and A/D Conversion is Enabled

When the DAADSCR.DAADST bit is 1 (measure against interference between D/A and A/D conversion is enabled), do not place the 12-bit A/D converter (unit 1) in the module stop state. It may halt D/A conversion in addition to A/D conversion.